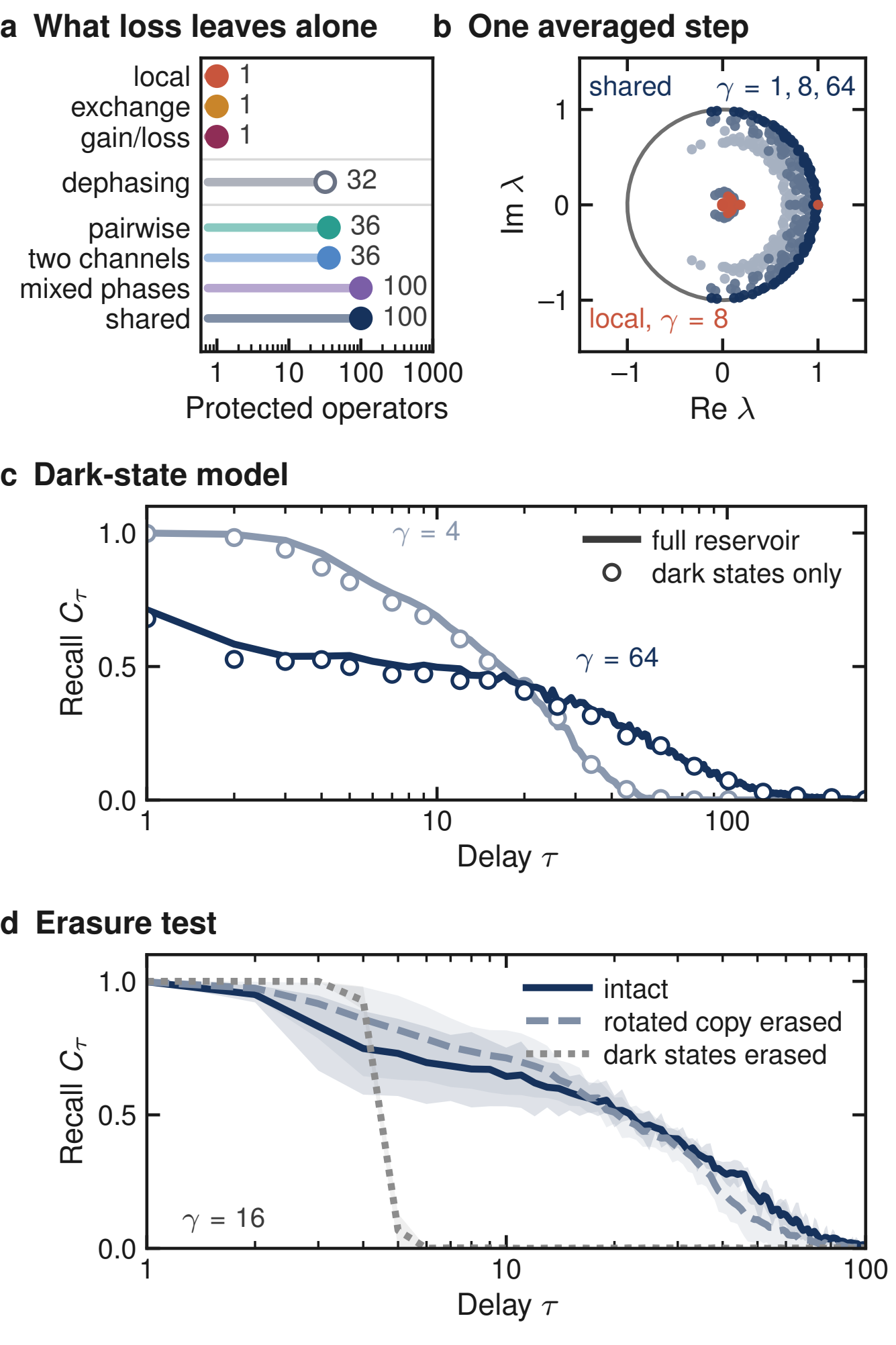

a What loss leaves alone
local
exchange
gain/loss
1
1
1
dephasing
32
pairwise
two channels
mixed phases
shared
36
36
100
100
1
10
100
1000
Protected operators
b One averaged step
shared
γ = 1, 8, 64
local, γ = 8
Im λ
Re λ
1
0
−1
c Dark-state model
Recall C_τ
Delay τ
γ = 4
γ = 64
full reservoir
dark states only
1.0
0.5
0.0
d Erasure test
intact
rotated copy erased
dark states erased
γ = 16